

DEPARTMENT OF MATHEMATICS

FEDERAL UNIVERSITY OF LAJIA

B.Sc. DEGREE First Semester Examination 2021/2022 Session

Course Code: MTH 111 Course Title: General Mathematics I

Instruction: Answer all Questions. Circle the correct answer Time Allowed: 2 hours. Units: 3

DEPARTMENT _____

Matric No. _____

1. The value of i (iota) is
 (a) -1 (b) 1 (c) $(-1)^{\frac{1}{2}}$ (d) $(-1)^{\frac{1}{4}}$
2. The maximum value and minimum value of $\sin \theta$ is...and...., respectively.
 (a) 1, -1 (b) 2, -1 (c) 3, -3 (d) 2, -2
3. Find the greatest common divisor of 36 and 54.
 (a) 2 (b) 18 (c) 3 (d) 6
4. A composite number that can be expressed as product of prime factors: 2, 3 and 5 is
 (a) 10 (b) 0 (c) 30 (d) 28
5. The matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ is a _____ matrix.
 (a) symmetric (b) non-singular (c) skew-symmetric (d) singular
6. What is the multiplicative inverse of a complex number Z ?
 (a) $\frac{\bar{Z}}{|Z|^2}$ (b) $\bar{Z}$ (c) $Z\bar{Z}$ (d) Z^{-1}
7. If $z = \frac{\sqrt{3}}{2} + \frac{i^5}{2} + \frac{\sqrt{3}}{2} - \frac{i^5}{2}$, then
 (a) $\operatorname{Re}(z) = 0$ (b) $\operatorname{Im}(z) = 0$ (c) $\operatorname{Re}(z) > 0$ (d) $\operatorname{Im}(z) < 0$
8. The ratio of the coefficient of x^{15} to the term independent of x in the expansion of $(x^2 + \frac{2}{x})^{10}$ is
 (a) 12:32 (b) 32:12 (c) 1:32 (d) 32:1
9. If $|A| = 0$ and $(\operatorname{Adj} A)B = 0$, then the system of equations has _____ solution(s).
 (a) exactly one (b) no (c) $n \times m$ (d) infinitely many
10. If $|A| \neq 0$, then the system of equations is consistent and has a unique solution, given by _____
 (a) $X = A^{-1}B$ (b) $X = AB$ (c) $X = A^T B$ (d) $X = |A|B$
11. Converting 15° into radians gives
 (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{12}$ (c) $\frac{\pi}{2}$ (d) $\frac{\sqrt{3}\pi}{2}$
12. The coefficient of x^p and x^q (p and q are positive integers) in the expansion of $(1+x)^{p+q}$ are
 (a) equal with opposite signs (b) reciprocal of each other
 (c) equal (d) none of these
13. If z represents a complex number then $\arg(z) + \arg(\bar{z})$ is
 (a) $\frac{\pi}{4}$ (b) 0 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$
14. If the amplitude of a complex number is $\frac{\pi}{2}$ then the number is
 (a) purely imaginary (b) purely real
 (c) 0 (d) neither real nor imaginary

15. The value of $i + i^{22} + i^{23} + i^{24} + i^{25}$ is
 (a) i (b) 1 (c) $-i$ (d) -1
16. The volume generated by rotating the triangle with vertices at $(0, 0)$, $(3, 0)$ and $(3, 3)$ about x-axis is
 (a) 2π (b) 36π (c) 18π (d) 9π
17. In how many ways can we select 6 people out of 10, of which a particular person is not included?
 (a) 9C_6 (b) ${}^{10}C_6$ (c) 9C_5 (d) ${}^{10}C_5$
18. If $\sin \theta = \frac{21}{29}$, what is the value of $\sec \theta + \tan \theta$, where θ lies in the first quadrant.
 (a) $\frac{5}{2}$ (b) $\frac{21}{20}$ (c) $2\frac{1}{2}$ (d) a and c
19. $\sin 2A = 2 \sin A$ is true when $A =$
 (a) 0° (b) 30° (c) 45° (d) 60°
20. Insert the five arithmetic means between 8 and 26.
 (a) 11, 12, 14, 15, 20 (b) 11, 14, 17, 20, 23 (c) 11, 13, 15, 17, 19 (d) 9, 11, 15, 18, 21
21. One of these is not a basic property of real numbers.
 (a) Algebraic property (b) Order property
 (c) Dissection property (d) Completeness property
22. If $a = 3 + i$ and $z = 2 - 3i$ then the points on the Argand diagram representing az , $3az$ and $-az$ are
 (a) Vertices of an equilateral triangle (b) Vertices of an isosceles triangle
 (c) Collinear (d) Vertices of a right angled triangle
23. Given two matrices S and T with orders (4×5) and (5×6) respectively. The matrix ST has order....
 (a) (4×5) (b) (4×6) (c) (5×5) (d) (5×6)
24. If $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, what is R^{-1} .
 (a) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (c) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (d) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
25. A matrix A is skew symmetric if
 (a) $A = -A$ (b) $A' = -A$ (c) $A = -A^T$ (d) $A^T = -A^T$
26. Solve the equation $\cos x + \sin x = \frac{1}{\cos x - \sin x}$ for values of x such that $0 \leq x \leq 2\pi$.
 (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{3}, \frac{2\pi}{3}$ (c) $0, \pi$ (d) $0, \frac{\pi}{2}$
27. Three consecutive terms in a geometric sequence are $n - 2$, n and $n + 3$. What is the common ratio of the sequence?
 (a) $\frac{3}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$
28. If the 10^{th} term of an AP is double the 3^{rd} term, what is the relationship between the first term and the common ratio?
 (a) $2a + 11d = 0$ (b) $a - 5d = 0$ (c) $a + 5d = 0$ (d) $2a - 11d = 0$
29. Given $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, evaluate $f(A) = A^2 - 4A + 3I$.
 (a) $\begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}$ (b) $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ (c) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (d) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
30. What will be added to $2x^2 + 5x$ to make it a perfect square.
 (a) $\frac{25}{16}$ (b) $\frac{19}{4}$ (c) $\frac{25}{4}$ (d) $\frac{25}{16}$

31. If $\cos \theta = \frac{12}{13}$, what is the value of $\sin \theta$?
 (a) $\frac{5}{13}$ (b) $\frac{13}{5}$ (c) $\frac{6}{13}$ (d) $\frac{13}{6}$
32. The minor of the entry, a_{23} of a given matrix is -14 . What is the value of the cofactor of the entry?
 (a) -14 (b) 14 (c) 13 (d) 15
33. The determinant of $P = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix}$ is ...
 (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{4}$ (d) 0
34. Let $A = \{t : t = 2n + 1, n \in \mathbb{N}\}$ and $B = \{0 < x < 10, x \in \mathbb{N}\}$. Find $A' \cap B'$.
 (a) $\{11, 13, 15, \dots\}$ (b) $\{3, 5, 6, \dots\}$ (c) $\{10, 11, 12, \dots\}$ (d) $\{10, 12, 14, \dots\}$
35. If α and β are the roots of the equation $3x^2 - 5x + 9 = 0$. Find the value of $\alpha^2 + \beta^2$.
 (a) $-\frac{29}{9}$ (b) $\frac{29}{9}$ (c) $\frac{11}{3}$ (d) $\frac{9}{29}$
36. Construct a quadratic equation if the sum and product of roots are 10 and 6, respectively.
 (a) $x^2 + 10x - 6 = 0$ (b) $x^2 - 10x - 6 = 0$ (c) $x^2 - 10x + 6 = 0$ (d) $x^2 - 5x - 9 = 0$
37. Find the conjugate of $(2 + i)^2$.
 (a) $3 + 4i$ (b) $3 + 5i$ (c) $3 - 2i$ (d) $5 + 3i$
38. For what value of x and y is $5x + 5yi$ and $10 + 18i$ are equal?
 (a) $(-2, -3)$ (b) $(4, 5)$ (c) $(-4, -5)$ (d) $(2, 3)$
39. How many combinations of 4 objects are there taken 2 at a time?
 (a) 6 (b) 7 (c) 8 (d) 3
40. In a club, there are 7 women and 5 men. A committee of 3 women and 2 men is to be chosen. How many different possibilities are there?
 (a) 250 (b) 350 (c) 150 (d) 130
41. Find the value of $\cos(45^\circ - 30^\circ)$.
 (a) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (b) $\frac{1-\sqrt{3}}{3}$ (c) $\frac{\sqrt{3}(1+\sqrt{3})}{4}$ (d) $\frac{\sqrt{3}-\sqrt{2}}{6}$
42. How many permutations are there in the word MISSISSIPPI?
 (a) $\frac{11!}{4!5!3!}$ (b) $\frac{11!}{4!3!3!}$ (c) $\frac{11!}{6!4!3!}$ (d) $\frac{11!}{4!4!3!}$
43. Find the fifth term in the expansion of $(1 - \frac{2}{3}x^3)$.
 (a) ${}_6C_4 (-\frac{2}{3}x^3)^4$ (b) ${}_6C_5 (-\frac{2}{3}x^3)^4$ (c) ${}_6C_6 (-\frac{2}{3}x^3)^4$ (d) $(-\frac{2}{3}x^3)^4$
44. What is the value $p(2)$ if $p(n) = \frac{n}{2n+1}$.
 (a) $\frac{2}{3}$ (b) $\frac{2}{5}$ (c) $\frac{2}{2n}$ (d) $-\frac{2}{3}$
45. The principle of mathematical induction involve the basic step and the _____ step.
 (a) deductive (b) constructive (c) inductive (d) selective
46. For any given set A , the family of all subsets of A is a well defined family of sets known as the _____ of A .
 (a) subset (b) empty set (c) cardinality (d) power set